the number of documents containing the term and then taking the logarithm of that quotient:

IDF = log(Total number of documents ÷ Number of documents containing the word)

In this example, the corpus is a small collection of eight documents, and *blue* appears in four of these documents.

IDF = log(8 ÷ 4)

TF–IDF CALCULATION

Finally, the TF–IDF score for *blue* in our sample sentence is calculated by multiplying the TF and the IDF scores:

TF–IDF = TF × IDF

Let's compute the actual values for TF–IDF for the word *blue* using the example provided; first, the term frequency (how often the word occurs in the document) is computed as follows:

TF = 1 ÷ 6

Assuming the base of the logarithm is 10 (commonly used), the inverse document frequency is computed as follows:

IDF = log10 (8 ÷ 4)

Now let's calculate the exact TF–IDF value for the word *blue* in the sentence, "The sky is blue and beautiful":

The Term Frequency (TF) is approximately 0.1670.

The Inverse Document Frequency (IDF) is approximately 0.301.

Thus, the TF–IDF (TF × IDF) score for *blue* is approximately 0.050.